FROM A DIMENSION-UNIFORM FOURIER THEOREM TO A FORMAL THREE-DIMENSIONAL PERIODIC GAUSSIAN LIMIT: THE RAW-LIFT PICTURE IN DIMENSIONS TWO, THREE, AND FOUR

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Abstract. For a massive Gaussian field on a periodic lattice at fixed physical volume, a dimension-uniform Fourier argument yields weak convergence of the unmodified atomic lift in every negative Sobolev carrier H^{-s} with $s > d/2$. We give that analytic theorem and center the formalization discussion on its three-dimensional instance. The Lean endpoint uses the actual three-dimensional finite-source law and the a^3 raw lift, and proves whole-sequence weak convergence on a weighted Fourier H^{-4} carrier to an explicit real-supported periodic Gaussian law. Its proof closes the concrete moment, compactness, joint Fourier-marginal, support, and uniqueness steps; the target law is constructed by restricting the existing four-dimensional Gaussian series to the fourth-frequency-zero slice and rescaling it. This target construction does not replace the three-dimensional lattice source.

We compare this result with the earlier two-dimensional H^{-3} and four-dimensional H^{-4} Lean theorems. The comparison distinguishes a paper proof uniform in dimension from dimension-specific Lean declarations, and it records the different target constructions and measure-identification arguments. In particular, the general-dimensional statement is a theorem proved on paper, not a Lean axiom or an arbitrary-dimensional Lean theorem. The three-dimensional formalization is evidence for one instance, not a proof of the general statement.

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1. Introduction

A fixed-volume lattice-to-continuum limit for a Gaussian field has two distinct layers. Fourier diagonalization gives the finite-dimensional covariance limits; a distribution-valued theorem additionally requires a common measurable state space, tightness of the full sequence, and uniqueness of every subsequential limit. For raw atomic lifts the first obstacle is already visible in the regularity: their Fourier coefficients repeat periodically and generally do not decay at high frequency.

This article connects three formal developments in a single model. The two-dimensional Lean theorem gives whole-sequence convergence on a Fourier realization of H^{-3}, with an existential target measure identified by its finite Fourier cylinders [4]. The four-dimensional theorem converges to a named Gaussian-series law on H^{-4} [5]. The three-dimensional theorem studied here uses the actual three-dimensional source and raw lift, proves convergence on H^{-4}, and constructs its target by a spectral restriction of the four-dimensional target [6]. These are three separately verified instances; no dimension change in one Lean declaration is being treated as an automatic derivation of another.

The mathematical background is a dimension-uniform paper theorem: for every integer $d \geq 1$, positive physical side length L, positive mass m, and $s > d/2$, the raw atomic laws converge on a weighted Fourier realization of $H^{-s}(\mathbb{T}_L^d)$. We include the proof because it explains why the three instances share the same probabilistic architecture. The formalized 3D endpoint is weaker in topology than the theorem's H^{-3} specialization: the released Lean theorem is in H^{-4}. It proves that concrete statement directly, with an H^{-3} moment bound supplying tightness in H^{-4}.

1.1. Terminology and claims.

We call the dimension-parameterized assertion a *general-dimensional theorem*, not an axiom. It has an analytic proof in this article but no single arbitrary-dimensional Lean declaration in the cited source release. The 3D Lean theorem is a compiled theorem for $d = 3$; it neither proves a theorem for all d nor validates an unspecified arbitrary-dimensional premise. This distinction is material to the formalization claim.

Our contribution is a formalization case study and a cross-dimensional comparison, not a priority claim for the classical Gaussian continuum limit. General-dimensional discrete-to-continuous results for other Gaussian models are known; for example, compact-manifold DGFF approximations and polyharmonic fields on tori have been studied in broad dimension ranges [2, 3]. Their covariance, interpolation or grounding conventions differ from the positive-mass periodic resolvent and raw atomic lift considered here.

The paper proceeds as follows. Section 2 fixes the normalization and derives the exact alias covariance. Section 3 proves the dimension-uniform paper theorem. Section 4 states the actual 3D Lean result and follows its proof chain. Section 5 compares the 2D, 3D, and 4D realizations. Section 6 records the formal evidence, and Section 7 summarizes claims and limitations.

2. Model and Fourier normalization

Fix an integer $d \geq 1$, a physical side length $L > 0$, a mass $m > 0$, and a positive integer N. Let $a = L/N$ and $\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^d$. For a real lattice configuration u, define

$$(\Delta_a u)(x) = a^{-2} \sum_{j=1}^{d} \big(u(x + e_j) + u(x - e_j) - 2u(x)\big).$$

The counting inner product gives

$$\sum_x u(x)(-\Delta_a u)(x) = a^{-2} \sum_{j,x} (u(x + e_j) - u(x))^2 \geq 0.$$

Thus $Q_N = m^2 - \Delta_a$ is positive definite. Write $G_N = Q_N^{-1}$ and let ϕ_N be the centered real Gaussian configuration with site covariance $a^{-d}G_N$. Its raw atomic lift is

$$X_N = a^d \sum_{x \in \Lambda_N} \phi_N(x)\, \delta_{[ax]}. \tag{1}$$

The factor a^{-d} belongs to the site covariance; the factor a^d belongs to the lift. They are distinct parts of the normalization.

Characters diagonalize Q_N. For a residue $p \in \Lambda_N$, its eigenvalue is

$$\lambda_N(p) = m^2 + 4a^{-2} \sum_{j=1}^{d} \sin^2(\pi p_j/N).$$

For $k \in \mathbb{Z}^d$, write $p_N(k)$ for its residue and define the physical Fourier coefficient

$$\widehat{X}_N(k) = a^d \sum_{x \in \Lambda_N} \phi_N(x) e^{-2\pi i k \cdot x/N}.$$

The continuum symbol is

$$\lambda(k) = m^2 + (2\pi/L)^2 |k|_2^2.$$

Lemma 2.1 (Exact alias covariance)**.** *For all $k, \ell \in \mathbb{Z}^d$,*

$$\mathbb{E}[\widehat{X}_N(k)\overline{\widehat{X}_N(\ell)}] = \frac{L^d\, \mathbf{1}_{p_N(k)=p_N(\ell)}}{\lambda_N(p_N(k))},$$
$$\mathbb{E}[\widehat{X}_N(k)\widehat{X}_N(\ell)] = \frac{L^d\, \mathbf{1}_{p_N(k)=-p_N(\ell)}}{\lambda_N(p_N(k))}.$$

In particular, $\mathbb{E}|\widehat{X}_N(k)|^2 \le L^d/m^2$ uniformly in N and k.

Proof. The inverse kernel is

$$G_N(x,y) = N^{-d} \sum_{p \in \Lambda_N} \lambda_N(p)^{-1} e^{2\pi i p \cdot (x-y)/N}.$$

Substitution in the covariance of the two sums yields a character sum equal to N^d exactly when the corresponding frequencies agree modulo N, and zero otherwise. The prefactor is

$$a^{2d} a^{-d} N^{-d} N^{2d} = (aN)^d = L^d.$$

Complex conjugation reverses the frequency and gives the second identity. Since $\lambda_N \ge m^2$, the uniform bound follows. □

The formula includes the zero mode and every finite-mesh alias. For even N, residues with each coordinate equal to 0 or $N/2$ are self-opposite; their sine coordinate vanishes. Thus one must not model all finite-mesh complex modes as distinct independent Gaussians. For any fixed finite set of continuum labels, all unintended residue collisions disappear once N is sufficiently large.

For fixed k, the elementary inequality $0 \le t^2 - \sin^2 t \le t^4/3$ gives

$$0 \le \lambda(k) - \lambda_N(p_N(k)) \le \frac{4\pi^4 a^2}{3L^4} \sum_{j=1}^{d} k_j^4.$$

Hence $\lambda_N(p_N(k)) \to \lambda(k)$. The physical factor L^{-4} in this error estimate is essential; it follows by substituting $t = \pi a k_j/L$ before multiplying by $4a^{-2}$.

For a real smooth periodic test function f, set $Ff(k) = \int_{\mathbb{T}_L^d} f(x)e^{-2\pi ik\cdot x/L}\,\mathrm{d}x$. The target field has covariance

$$Q(f,g) = L^{-d}\sum_{k\in\mathbb{Z}^d} \frac{\overline{Ff(k)}Fg(k)}{\lambda(k)}. \tag{2}$$

Equivalently, its physical Fourier coefficients satisfy $\mathbb{E}[\widehat{Z}(k)\overline{\widehat{Z}(\ell)}] = L^d\mathbf{1}_{k=\ell}/\lambda(k)$.

3. The dimension-uniform paper theorem

For $k \in \mathbb{Z}^d$ let $R(k) = \max_j |k_j|$, and define $w_t(k) = (1+R(k))^{-2t}$. The weighted Fourier carrier $\mathcal{H}^{-t}$ consists of coefficient families z with $z(-k) = \overline{z(k)}$ and

$$\|z\|_{-t}^2 = \sum_{k\in\mathbb{Z}^d} w_t(k)|z(k)|^2 < \infty.$$

It is a separable real Hilbert space, realized as the closed conjugate-symmetric subspace of a weighted ℓ^2 space. Its topology is equivalent to the usual Fourier Sobolev topology, since $R(k)^2 \le |k|_2^2 \le dR(k)^2$.

Theorem 3.1 (Fixed-volume raw-lift limit in dimension d)**.** *For each integer $d \ge 1$, $L > 0$, $m > 0$, and $s > d/2$, the laws of the actual lattice fields X_N in* (1) *converge weakly on $\mathcal{H}^{-s}(\mathbb{T}_L^d)$ to the centered real periodic Gaussian law with covariance* (2)*.*

Proof. Choose r with $d/2 < r < s$. The number of lattice frequencies with $R(k) = q$ is

$$(2q+1)^d - (2q-1)^d = O_d(q^{d-1}),$$

so $S_{d,r} = \sum_k w_r(k) < \infty$. Lemma 2.1 and Tonelli's theorem imply

$$\sup_N \mathbb{E}\|X_N\|_{-r}^2 \le L^d m^{-2} S_{d,r} < \infty.$$

A raw finite atomic source has periodic Fourier coefficients and belongs to $\mathcal{H}^{-r}$ precisely on the safe side $r > d/2$. For $s > r$, the $\mathcal{H}^{-r}$-ball of radius B has uniformly small $\mathcal{H}^{-s}$-tails:

$$\|z - P_M z\|_{-s}^2 \le (1+M)^{-2(s-r)}B^2,$$

where P_M retains $R(k) \le M$. The truncation is finite-dimensional, hence the ball has compact closure in $\mathcal{H}^{-s}$. Markov's inequality gives tightness of the actual laws there.

Construct the target explicitly. Choose one representative from each pair $\{k,-k\}$ of nonzero modes. Let ξ_0 and ξ_k, η_k be independent standard real Gaussians and set

$$Z(0) = L^{d/2}m^{-1}\xi_0, \qquad Z(k) = \frac{L^{d/2}}{\sqrt{2\lambda(k)}}(\xi_k + i\eta_k), \qquad Z(-k) = \overline{Z(k)}.$$

Then $\mathbb{E}\sum_k w_s(k)|Z(k)|^2 = L^d\sum_k w_s(k)/\lambda(k) < \infty$, so this is a measurable random element of the real carrier. Its covariance is (2).

For any fixed finite family of real and imaginary Fourier evaluations, both the source and target vectors are centered Gaussian. Their covariance matrices converge by the exact alias identities and the fixed-mode convergence $\lambda_N(p_N(k)) \to \lambda(k)$. Thus their joint laws converge. Tightness gives a weakly convergent subsubsequence from every subsequence; continuous Fourier projections pass its finite-dimensional laws to the limit. The real Fourier coordinates determine Borel probability measures on the separable carrier, so every subsequential limit is the law of Z. Tightness together with uniqueness of subsequential limits yields convergence of the full sequence. □

Evidence boundary. This is an analytic proof in the article. It is not one generic Lean declaration in the cited source release. The dimension-specific formalization results are stated separately in Section 4.

The distinction between the raw approximants and the limit matters. The limiting Gaussian has finite expected $\mathcal{H}^{-s}$ norm already for $s > d/2 - 1$, by the high-frequency factor $\lambda(k)^{-1} \asymp |k|^{-2}$. A nonzero finite raw lift, however, has a periodic nondecaying Fourier subsequence and generally fails to belong to $\mathcal{H}^{-s}$ when $s \le d/2$. Smoother convergence statements therefore require a different approximation, such as interpolation or spectral truncation.

4. The formal three-dimensional instance

Set $d = 3$, $N = n + 1$, and $a = L/N$. The finite source is the centered Gaussian on $(\mathbb{Z}/N\mathbb{Z})^3$ with site covariance $a^{-3}(m^2 - \Delta_a)^{-1}$. The Lean development maps this actual finite law by the physical raw lift

$$X_N = a^3 \sum_{x \in (\mathbb{Z}/N\mathbb{Z})^3} \phi_N(x)\delta_{[ax]}.$$

The exported endpoint is

> `Pphi2Clean.Continuum.periodic3DFixedVolume_weak_tendsto_gaussianTarget`.

It assumes only $L > 0$ and $m > 0$ and proves convergence of the full sequence of probability measures on the weighted Fourier H^{-4} carrier to the named target measure. No caller-supplied tightness, marginal convergence, target identification, or subsequential uniqueness premise is part of its theorem type.

4.1. **The target and the normalization.** The target is a three-dimensional periodic-resolvent Gaussian law with symbol

$$\lambda_3(k) = m^2 + (2\pi/L)^2(k_1^2 + k_2^2 + k_3^2).$$

The Lean construction reuses the four-dimensional Gaussian Fourier carrier only to construct this target: restrict its frequencies to $(k_1, k_2, k_3, 0)$ and multiply the restricted field by $(\sqrt{L})^{-1}$. The squared scale contributes L^{-1}, so the four-dimensional factor L^4 becomes L^3, while the restricted mass symbol is exactly $\lambda_3(k)$. This is a spectral restriction, not a spatial trace.

The real zero coordinate has variance L^3/m^2; its imaginary coordinate vanishes. For each nonzero canonical mode, the real cosine and sine coordinates have variance $L^3/(2\lambda_3(k))$. These are joint Gaussian statements: equality of coordinate variances alone would not identify a finite-dimensional law. The formal development also proves the cross-covariance identities and treats repeated and degenerate coordinates.

4.2. **Proof chain.** The machine-checked route has the following concrete stages.

(1) **Actual source and lift.** The finite-source map is defined from the $d = 3$ lattice Gaussian law. Its raw atomic Fourier coefficient is the a^3-weighted site sum, and the weighted carrier map is shown to represent that source.
(2) **Moment control and tightness.** A uniform second-moment estimate is proved in H^{-3}. Closed H^{-3} energy balls have compact closure in the H^{-4} carrier by finite-frequency truncation. Markov's inequality and Prokhorov's theorem give subsequential weak limits of the actual source laws.
(3) **Finite real Fourier marginals.** Every fixed finite family of actual source cosine/sine evaluations has a centered Gaussian law. The joint covariance converges to the $d = 3$ periodic-resolvent covariance. The proof includes repeated modes and eventual exclusion of unintended aliases and self-negative Nyquist modes.

(4) **Explicit target marginals.** The target's supported canonical projections are shown to be jointly Gaussian with the same resolvent covariance. The projection law is identified with the corresponding finite-dimensional Gaussian, not merely coordinatewise matched.
(5) **Real support and uniqueness.** Source and target lie on the closed conjugate-symmetric real carrier. This support passes to weak limits. On that real carrier, the canonical coordinates determine the Borel probability measure; hence every subsequential limit is the named target, and tightness yields full-sequence convergence.

The Lean endpoint is in H^{-4}, whereas the analytic theorem of Section 3 allows $s = 3$ in dimension three. In its general proof one may choose $(r, s) = (2, 3)$; the current Lean proof instead controls an H^{-3} moment and concludes in H^{-4}, corresponding to $(r, s) = (3, 4)$. Thus the formal result is a compatible but weaker-topology instance, proved by its own carrier-level chain. It does not claim a Lean proof of the sharper H^{-3} specialization.

Evidence boundary. The final Lean theorem's reported axiom footprint is exactly `[propext, Classical.choice,Quot.sound]`. This is the ordinary Lean/Mathlib foundational footprint; it is not a claim that the proof has no foundational axioms. The release commit and theorem transcript are cited in [6].

5. How the three dimensions fit together

The analytic theorem in Section 3 provides a common mathematical route, but the three native Lean endpoints remain distinct declarations. Table 1 separates the source normalization, carrier choice, target construction, and identification method.

Table 1. Native periodic Gaussian weak-limit theorems in dimensions two, three, and four.

Feature	Two dimensions	Three dimensions	Four dimensions
Site covariance	$a^{-2}G_N$	$a^{-3}G_N$	$a^{-4}G_N$
Raw lift	$a^2 \sum \phi_x \delta_{ax}$	$a^3 \sum \phi_x \delta_{ax}$	$a^4 \sum \phi_x \delta_{ax}$
Energy / limit carrier	H^{-2}/H^{-3}	H^{-3}/H^{-4}	H^{-3}/H^{-4}
Target	Cylinder law	Restricted 4D Gaussian	Gaussian series
Identification	Fourier cylinders	Joint real modes	Real modes and support
Source law	2D lattice	Actual 3D source	4D lattice

The two-dimensional theorem returns a probability measure on its concrete completion and characterizes its finite Fourier cylinders by the periodic-resolvent marginals; it does not package the endpoint as a named Gaussian-series measure. The four-dimensional theorem constructs such a target explicitly and handles uniqueness by canonical real coordinates together with conjugate-symmetric support. The three-dimensional formalization follows the latter style of explicit target identification but has a different source: the finite lattice source, physical lift, and all covariance limits are genuinely three-dimensional. Only the target construction reuses the fourth-frequency-zero slice of the 4D Gaussian carrier.

The public Lean endpoints compared here are

2D: `Pphi2Clean.Continuum.periodicHMinusThree_fixedVolume_weak_limit_exists_with_expected_cylinders`
3D: `Pphi2Clean.Continuum.periodic3DFixedVolume_weak_tendsto_gaussianTarget`
4D: `Pphi2Clean.Continuum.fs4PeriodicFixedVolume_weak_tendsto_free`

The common proof architecture is

$$\begin{aligned}&\text{finite source covariance and raw lift}\\ &\longrightarrow \text{uniform stronger-space moment}\\ &\longrightarrow \text{tightness in a weaker carrier}\\ &\longrightarrow \text{joint finite-coordinate convergence}\\ &\longrightarrow \text{uniqueness of subsequential limits}\\ &\longrightarrow \text{full-sequence convergence.}\end{aligned}$$

What is not shared is equally important: the summability and compact embedding are instantiated at different exponents; the 2D target is expressed by cylinders; and the 3D/4D explicit-target arguments must account for support in the real carrier. The 3D theorem therefore strengthens the project comparison by supplying a third actual source dimension, not by turning the three formal developments into one parameterized Lean theorem.

5.1. **Relation to earlier continuum-limit work.** The continuum Gaussian model and the compactness-plus-finite-marginal strategy are classical. Cipriani and van Ginkel prove compact-manifold DGFF limits in arbitrary dimension using graph approximations and a strong Sobolev topology [2]. Dello Schiavo, Herry, Kopfer, and Sturm prove discrete-to-continuous results for dimension-dependent polyharmonic fields on tori [3]. Those theorems differ in covariance, mass/zero-mode conventions, and approximation map. We do not claim that a three-dimensional Gaussian continuum limit is new in probability theory. The narrower contribution is a kernel-checked formalization of this fixed-volume, positive-mass, cubic periodic raw-source instance, with a source and target contract that can be compared against the 2D and 4D releases.

6. Lean development and reproducibility

The 3D source release separates the finite lattice construction, weighted carrier, moment estimate, compact embedding, finite marginal convergence, target construction, real support, and weak-limit assembly into named Lean modules. The following table records representative interfaces rather than a line-count claim.

The 3D modules are published in the source snapshot [6]. The theorem guide records the exact endpoint and the proof chain. The following metadata refers to the complete pinned repository, which includes the previously formalized 2D and 4D developments; the counts are not attributed to the 3D proof alone. The package-wide source census was run on 2026-11-10 using `verify.py --count`.

The source census is the package-wide `verify.py --count` result, while the endpoint footprint comes from the separate joint `#print axioms` transcript in `audit/`. The repository manifest and census checks passed; a targeted build of the 2D, 4D, and 3D theorem modules and the joint contract replay also passed at this release. These are complementary checks: source cleanliness does not prove theorem meaning, and a kernel axiom query does not establish that the stated theorem is the intended continuum limit.

6.1. **What the formalization does not certify.** The 3D endpoint does not establish the arbitrary-d statement in Lean, nor does the 3D specialization establish that theorem for other dimensions. It does not prove convergence in H^{-3} at the formal endpoint, an interacting-field limit, an infinite-volume limit, Osterwalder–Schrader reconstruction, or a spatial trace theorem. It also does not claim a separately packaged Lean isomorphism between the weighted Fourier carrier and a pre-existing Bessel-potential distribution type. These scope limits are part of the theorem contract.

TABLE 2. Representative components of the 3D formalization.

Lean module	Role in the theorem
`Periodic3DFiniteSourceLift.lean`	Actual 3D source law and physical raw atomic lift
`Periodic3DCarrierEnergy.lean`	H^{-3} moment identity and uniform energy control
`Periodic3DCompactEmbedding.lean`	Compact inclusion from the stronger energy space into H^{-4}
`Periodic3DTightness.lean`	Markov tightness and subsequence extraction for the actual laws
`Periodic3DFiniteMarginal.lean`	Joint covariance convergence for finite real Fourier families
`Periodic3DGaussianTarget.lean`	Explicit target from the 4D Gaussian Fourier slice
`Periodic3DRealCarrier.lean`	Closed conjugate-symmetric support for source and target
`Periodic3DGaussianWeakLimit.lean`	Limit identification and full-sequence weak convergence

TABLE 3. Pinned source and formal-contract metadata.

Item	Verified value
Release commit	`b0be3960210d5c638513de9995850e318b1a5b4d`
Lean toolchain	`leanprover/lean4:v4.34.0-rc2`
Mathlib revision	`852d20a84581aeba8527bd01d0be361d8c0e71c0`
Source census	116 modules; 22,021 lines; 1,308 declarations, of which 939 are theorem or lemma forms
Source holes	0 `sorry`; 0 declared `axiom` in the pinned package
Joint endpoint replay	3 signatures and typed examples; exit status 0
Endpoint axiom output	All three endpoints report `propext`, `Classical.choice`, and `Quot.sound`

The formalization is useful because normalization-sensitive facts are proved together with the probability argument: the a^3 source lift, the L^3 target covariance, zero-mode handling, joint covariance of finite families, Nyquist eventual separation, and support-aware uniqueness. The mathematical reader can inspect these declarations independently of the article's analytic proof.

7. Evidence ledger and limitations

Table 4 records what supports the principal claims and where the formal evidence stops.

The arbitrary-dimensional theorem is proved analytically by the moment, compactness, finite-marginal, and uniqueness arguments above. The source release separately verifies the 2D, 3D, and 4D endpoint declarations. These statements must not be collapsed into an assertion that the general theorem has been formalized. Likewise, the standard Lean axiom footprint is evidence about the elaborated theorem's dependencies, not an assertion that formal proof removes foundational axioms.

The present article makes no priority claim for Gaussian lattice scaling limits. It is a formalization-centered report of a concrete three-dimensional source-to-target theorem and a comparison of three dimension-specific Lean proof chains. Neither the construction of the 3D

TABLE 4. Claim-to-evidence ledger.

Claim	Evidence status	Boundary
General-d limit, $s > d/2$	CONFIRMED	Analytic proof; no generic Lean declaration
3D source limit on H^{-4}	CONFIRMED	Final theorem and axiom query at the pinned release
2D limit on H^{-3}	CONFIRMED	Separate theorem and carrier
4D limit on H^{-4}	CONFIRMED	Separate source; target construction only for 3D
3D target normalization	CONFIRMED	4D frequency slice scaled by $L^{-1/2}$
3D Lean limit on H^{-3}	UNVERIFIED	No such endpoint is claimed; formal result is H^{-4}
Arbitrary-d Lean convergence	UNVERIFIED	No generic declaration; paper proof is analytic
Interacting, infinite-volume, OS	OUT OF SCOPE	Requires separate hypotheses and arguments

target from a 4D Gaussian series nor the existence of three formalized dimensions proves an arbitrary-dimensional Lean theorem.

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